

Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

POSTLAB EXERCISES (20 points)*Submit the postlab to the TA at the end of the lab.***Measurement using Photogate Timers (4 points)****Question 3****1 point**

Using your measurements from Table 2, calculate the means $\bar{t}_1$ and $\bar{t}_2$ and the standard deviations Δt_1 and Δt_2 . Include units.

 $\bar{t}_1 =$ _____ $\bar{t}_2 =$ _____ $\Delta t_1 =$ _____ $\Delta t_2 =$ _____**Question 4****1 point**

Using eq. (12) with $\bar{t}_1$ and $\bar{t}_2$, find the average acceleration $\bar{a}$. Show all work and include units.

 $\bar{a} =$ _____

Question 5**2 point**

- (a) (1 point) Estimate your uncertainties in L (ΔL) and D (ΔD) based on how accurately you read the ruler. Use the error propagation formula below to estimate Δa , the uncertainty in the acceleration:

$$\Delta a = \bar{a} \cdot \sqrt{\left(\frac{\Delta L}{L}\right)^2 + \left(\frac{2\Delta D}{D}\right)^2 + \left(\frac{2\Delta t_1}{\bar{t}_1}\right)^2 + \left(\frac{2\Delta t_2}{\bar{t}_2}\right)^2}.$$

Include units.

$\Delta L =$ _____

$\Delta D =$ _____

$\Delta a =$ _____

- (b) ($1/2$ point) Which of the four terms in the square root contributed most to Δa ?

- (c) ($1/2$ point) What is one possible reason this term dominates the uncertainty?

Measurement using Manual Timers (8 points)**Question 6****1 point**

Calculate the mean and standard deviation of your hand timer data (Table 3). Include units.

$\bar{t} =$ _____

$\Delta t =$ _____

Question 7**1 point**

Using eq. (15) with $\bar{t}$, find the average acceleration $\bar{a}$. Show all work and include units.

$\bar{a} =$ _____

Question 8**2 point**

- (a) (1 point) Estimate your uncertainty in S (ΔS) based on how accurately you read the ruler. Use the error propagation formula below to estimate Δa :

$$\Delta a = \bar{a} \cdot \sqrt{\left(\frac{\Delta S}{S}\right)^2 + \left(\frac{2\Delta t}{\bar{t}}\right)^2}.$$

Include units.

$\Delta S =$ _____

$\Delta a =$ _____

- (b) (1 point) Which quantity, S or t , contributes more to the uncertainty Δa ?

Question 9**1 point**

Use eq. (2) to estimate g from both the photogate and manual timer data. Compute the uncertainty on g using

$$\Delta g = \frac{M_1 + M_2}{M_1 - M_2} \Delta a.$$

Include units.

$g_{\text{photogate}} =$ _____ $\pm$ _____

$g_{\text{hand timer}} =$ _____ $\pm$ _____

Question 10**3 point**

Compare the accuracy and precision of the two methods. The accepted value is $g = 9.8039 \text{ m s}^{-2}$.

(a) (1 point) Which method was more precise? Explain based on your estimates for g .

(b) (1 point) Which method was more accurate? Explain based on your estimates for g .

(c) (1 point) How many standard deviations apart are your two estimates for g ? Use the formula

$$N_{\text{SD}} = \left| \frac{g_{\text{photogate}} - g_{\text{hand timer}}}{\Delta g_{\text{photogate}} + \Delta g_{\text{hand timer}}} \right|.$$

Accounting for the Rotational Inertia of the Pulley (8 points)**Question 11****1 point**

Using the data in Table 4, calculate $(M_1 - M_2)/(M_1 + M_2)$ for each mass combination. Then compute the acceleration for each trial using eq. (15). Include units.

$\frac{M_1 - M_2}{M_1 + M_2}$	Acceleration		
	0 rings	1 ring	2 rings

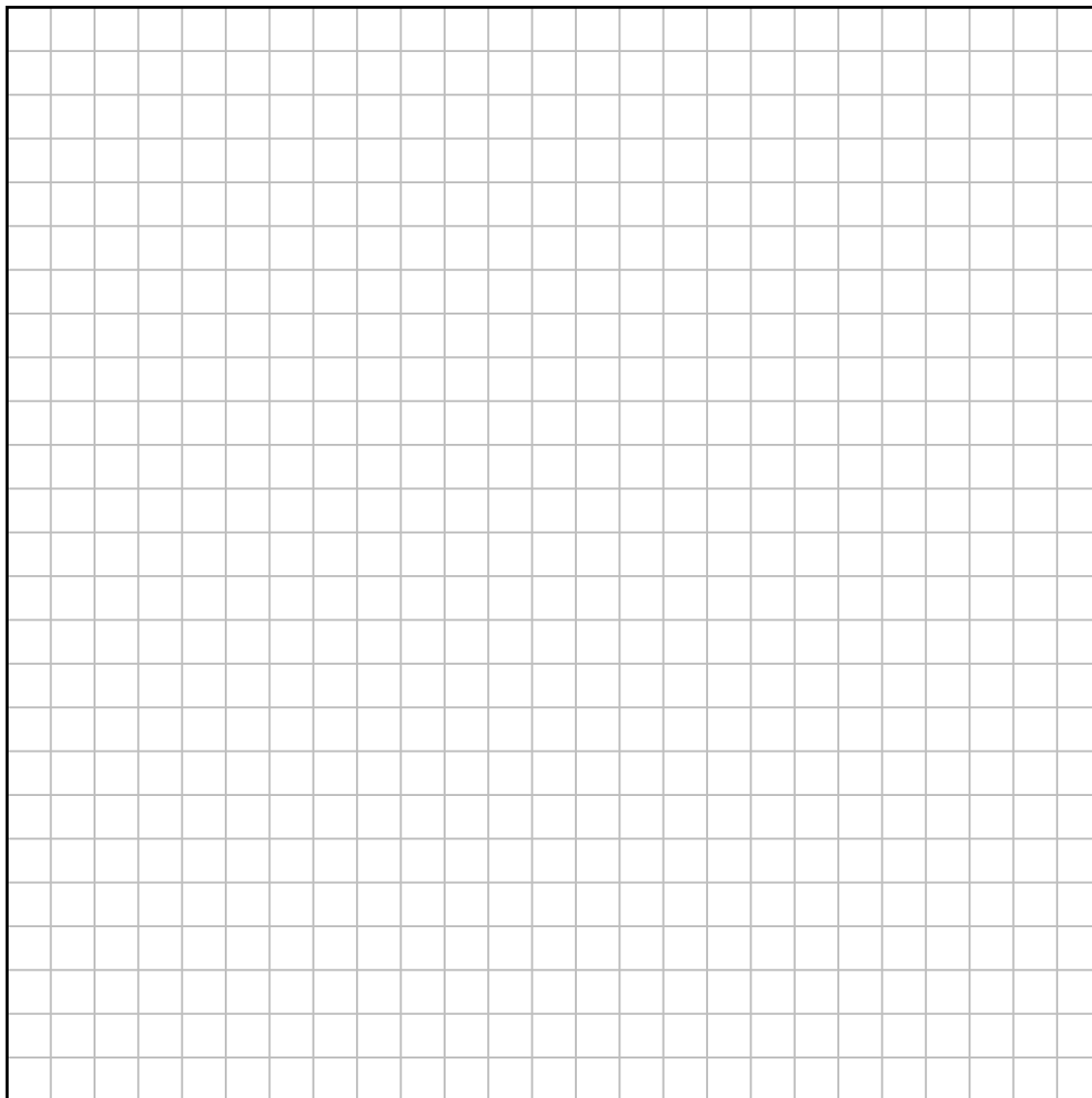
Question 12**2 points**

If the pulley is massless, eq. (1) holds and

$$a = \frac{M_1 - M_2}{M_1 + M_2}g.$$

A plot of a versus $(M_1 - M_2)/(M_1 + M_2)$ should then be a straight line with slope g . To assess whether the pulley mass affects the results, make this plot and compare the slope to the accepted value $g = 9.8039 \text{ m s}^{-2}$.

Plot a versus $(M_1 - M_2)/(M_1 + M_2)$ for the 0-ring, 1-ring, and 2-ring data on the grid below. Draw a best-fit line through each set of points. Include axis labels, units, a title, and a legend.



Question 13**1 point**

How does your measured value of g change as the pulley mass increases? *Hint: g is the slope of each best-fit line.*

Question 14**1 point**

Given your answer to question 13, can the effect of the pulley's mass explain the discrepancy between your estimates of g in question 9 and the accepted value $g = 9.8039 \text{ m s}^{-2}$?

Question 15**1 point**

What is another significant source of error in Atwood's machine that is not explored in this lab? How might it affect your results?

Question 16**1 point**

(a) ($1/2$ point) What is one improvement to Atwood's machine that would increase the accuracy of your results?

(b) ($1/2$ point) What is one improvement that would increase the precision?

Question 17**1 point**

Atwood's machine has several systematic errors that would not be present in a simpler experiment. What is one reason we might nonetheless expect better results from Atwood's machine than from simply dropping objects off tall buildings, as Galileo did?